



SACE STAGE 2 · REVISION CHECKLIST

# Mathematical Methods

Every topic you need to know, in one checklist.

6 topics

47 checkpoints

TopMark Tutors

## How to use this checklist

Work through every line and tick honestly. Do the whole thing in one sitting before you revise anything — that first pass becomes your study plan for the term.

1

### Revised

You have gone back over the content and understand it.

2

### Practised

You have done real exam questions on it, not just read notes.

3

### Confident

You could do it in the exam, under time, with no notes.

**Most students stop at column 1.** Re-reading notes feels like studying, but marks are won in columns 2 and 3. Anything ticked in column 1 and blank in column 3 is exactly where your marks are leaking.

*Externally assessed by written examination (30% of final grade).*

Produced by TopMark Tutors as a free study resource, structured around the SACE Stage 2 Mathematical Methods subject outline. It is a revision aid only and is not produced, endorsed, or reviewed by the SACE Board of South Australia. Always check the current subject outline for your enrolment year and confirm content with your teacher.



## Topic 1 — Further Differentiation and Applications

TOPIC 1 OF 6

## Differentiation rules

| CONTENT                                                                            | REVISED                  | PRACTISED                | CONFIDENT                |
|------------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Chain rule, product rule and quotient rule, including rules applied in combination | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Differentiating exponential functions ( $e^x$ and $a^x$ )                          | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Differentiating trigonometric functions (sin, cos, tan)                            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Second derivatives and their meaning                                               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## Applications

|                                                                                      |                          |                          |                          |
|--------------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Increasing/decreasing functions, stationary points and the nature of each            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Points of inflection and concavity using the second derivative                       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Curve sketching from first and second derivative information                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Optimisation problems, including forming the function from a worded context          | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Checking solutions against a restricted domain                                       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Rates of change, related rates and kinematics (displacement, velocity, acceleration) | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## WEAK SPOTS TO COME BACK TO

---

---

---

---



## Topic 2 — Discrete Random Variables

TOPIC 2 OF 6

## Discrete distributions

| CONTENT                                                                   | REVISED                  | PRACTISED                | CONFIDENT                |
|---------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Recognising a valid probability distribution                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Calculating $E(X)$ , $\text{Var}(X)$ and standard deviation               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Interpreting expected value in context (e.g. fair games, expected profit) | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Effect of linear transformation: $E(aX+b)$ and $\text{Var}(aX+b)$         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## The binomial distribution

|                                                                              |                          |                          |                          |
|------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Conditions required for a binomial distribution                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Calculating binomial probabilities, including 'at least' and 'at most' cases | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Mean and variance of a binomial distribution ( $np$ and $npq$ )              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## WEAK SPOTS TO COME BACK TO

---

---

---

---

---



## Topic 3 — Integral Calculus

TOPIC 3 OF 6

## Anti-differentiation

| CONTENT                                                                 | REVISED                  | PRACTISED                | CONFIDENT                |
|-------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Anti-derivatives of polynomial, exponential and trigonometric functions | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Finding the constant of integration from a given condition              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Integration by recognition of the chain rule (simple substitution)      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## Definite integrals and applications

|                                                                       |                          |                          |                          |
|-----------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Evaluating definite integrals and interpreting a negative value       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Area under a curve and area between two curves                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Determining where curves intersect to establish limits of integration | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Using a rate function to find total accumulated change                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The fundamental theorem of calculus                                   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Properties of definite integrals (splitting and combining intervals)  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## WEAK SPOTS TO COME BACK TO

---

---

---

---

---



## Topic 4 — Logarithmic Functions

TOPIC 4 OF 6

## Logarithm laws and equations

| CONTENT                                                             | REVISED                  | PRACTISED                | CONFIDENT                |
|---------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Index and logarithm laws, including the change of base rule         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Solving exponential equations using logarithms, giving exact values | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Solving logarithmic equations and checking for invalid solutions    | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## Calculus and modelling

|                                                                              |                          |                          |                          |
|------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Differentiating and integrating logarithmic and exponential functions        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Graphs of $y = \log_a x$ and $y = a^x$ , including asymptotes and intercepts | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Exponential growth and decay models, finding the rate constant $k$           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Interpreting the limitations of an exponential model in context              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## WEAK SPOTS TO COME BACK TO

---

---

---

---

---



## Topic 5 — Continuous Random Variables and the Normal Distribution

TOPIC 5 OF  
6

### Continuous random variables

| CONTENT                                                            | REVISED                  | PRACTISED                | CONFIDENT                |
|--------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Probability density functions and finding probabilities using area | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Mean and variance of a continuous random variable                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

### The normal distribution

|                                                                        |                          |                          |                          |
|------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Properties of the normal curve and the empirical (68–95–99.7) rule     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Standardising to z-scores and using them to compare values             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Finding probabilities for a given interval                             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Working backwards from a probability to find the corresponding x value | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Recognising when a normal model may not be appropriate                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

#### WEAK SPOTS TO COME BACK TO

---

---

---

---

---



## Topic 6 — Sampling and Confidence Intervals

TOPIC 6 OF 6

## Sampling

| CONTENT                                                       | REVISED                  | PRACTISED                | CONFIDENT                |
|---------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Population vs sample, and sources of bias in sampling         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The distribution of the sample mean and the sample proportion | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| The effect of sample size on variability                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## Confidence intervals

|                                                                      |                          |                          |                          |
|----------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| Calculating standard error for a proportion and for a mean           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Constructing confidence intervals and calculating margin of error    | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Correctly interpreting a confidence interval (the common exam trap)  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Effect of changing sample size or confidence level on interval width | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## WEAK SPOTS TO COME BACK TO

---

---

---

---

---



## WHAT TO DO WITH THE BLANKS

# The gaps you just found are the whole point.

Every line you could not tick in the 'Confident' column is a mark you are currently at risk of dropping. Our SACE Exam Crash Course is built to close exactly those gaps — we mapped four years of real SACE papers to find where marks are actually lost, then weighted the teaching toward those points instead of re-covering content you already know.

### The SACE Exam Crash Course — Mathematical Methods

- 28 September – 10 October · Methods, Chemistry, Physics, Biology
- 12 hours of teaching per subject · online or in person, same live sessions
- TopMark Mathematical Methods textbook included free
- Finishes with a full mock exam, marked and returned with your personal mark-loss breakdown
- 32 five-star Google reviews · 100% of our 2025 cohort scored above a 90 ATAR

From \$349 for one subject, or \$20/hr across all four. A \$99 deposit secures your spot. Early bird pricing ends 6 September.

[topmarktutors.com.au](https://topmarktutors.com.au) →

Call or text 0450 113 766 · Adelaide